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Education

MIT-WPU, (CGPA-7.9)

B.Tech Computer Science Engineering

Pune, Maharashtra, India

2022 – 2026

JKG International School

CBSE Class 10 & 12

Ghaziabad, UP, India

March 2020 & 22

Summary

Computer Science student specializing in **Generative AI and Computer Vision**, with hands-on experience building **LLM-powered systems, agentic workflows, scalable retrieval pipelines, and deep learning models**. Experienced in developing AI products using **RAG, semantic search, custom embeddings, and full-stack ML applications**.

Experience

Data Science Intern – Involead

July 2025 – December 2025

- Built multi-agent RAG, Graph-RAG, and hybrid retrieval pipelines, improving answer quality, enabling reasoning over structured and unstructured data, and supporting complex client-facing GenAI use cases.
- Developed scalable search and retrieval systems for 400K+ documents using semantic, sparse, and hybrid search approaches and improved retrieval performance through chunking, extraction, and indexing optimizations, reducing latency and increasing relevance.
- Delivered evaluation and modeling solutions, including an LLM evaluation framework for RAG analysis and a custom embedding model for semantic similarity, improving reliability and performance of production-oriented AI workflows.

Projects

Self-Learning RAG

- Increased RAG answer quality by **8.8%** and improved benchmark pass rates from **41% to 79%** by building an adaptive self-learning retrieval pipeline with query rewriting, feedback-driven optimization, and dynamic retrieval tuning
- Improved retrieval relevance by **24%** and reduced failure cases through automated Top-K overrides, rewrite pattern learning, and re-chunking triggers driven by continuous failure analysis
- Achieved **62% evaluation pass rates** across benchmark queries by developing a **FastAPI + MongoDB** feedback system for real-time metrics tracking and continuous improvement loops

CodeLens

- Architected MCP server with **FastAPI** exposing file system and LLM tools, reducing API complexity through standardized interfaces and local **Ollama** integration
- Engineered code review pipeline with **LLaMA 3.1 (8B)** using custom prompts, detecting security issues, anti-patterns, and performance bottlenecks with 85%+ accuracy
- Built **React** frontend with async backend integration, handling edge cases and achieving sub-10 seconds review time for 150-line codebases

VisuDocAI

- Built a full-stack document intelligence system with **Flask** and **React**, achieving **95%+ text extraction accuracy (CER 4.7%, WER 4.5%)** across PDFs and scanned images using **PyMuPDF** and **Tesseract OCR**
- Improved document understanding quality by implementing a multi-stage **LLM pipeline** (classification, Q&A, summarization) via **OpenRouter**, achieving **100% classification accuracy** and **62% ROUGE-1** for summarization using domain-specific models
- Delivered a reliable end-to-end system with **100% processing success rate** and interactive query interface, enabling real-time document analysis with async file handling

Footymind

- Achieved **98.6%** detection accuracy (mAP@0.5) by training custom **YOLOv5** model with **Hungarian Algorithm** tracking
- Built full video-to-event pipeline for automated football match analysis and event logging
- Engineered spatial analysis and possession logic to detect passes, tackles, and goals with frame-level accuracy

Skills

- **Relevant Courses:** Machine Learning, Computer Vision, Operating Systems, Computer Networks, Database and Management Systems, Object Oriented Programming
- **Programming Languages:** C, C++, Python, JavaScript
- **Web Development:** HTML, CSS, React, Node.js, Express, Flask
- **Frameworks & Libraries:** Tailwind CSS, Three.js, Framer Motion, MediaPipe
- **AI/ML:** YOLOv5, XGBoost, BERT, LSTM, Neural Networks, LLMs, SLMs, RAG, LangChain, LangGraph
- **Tools & Platforms:** Git, MongoDB, Agile methodologies, Ollama, MCP, OpenRouter, n8n, Zapier